

JAMIL RAFIQUE SHAIKH

Senior Consultant 1 - Senior SRE | Cloud-Native Platform Engineer | DevOps/SRE Consultant

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PROFESSIONAL SUMMARY

Site Reliability Engineer with 11+ years of experience across Linux systems, DevOps, Kubernetes, cloud platforms, CI/CD, observability, incident management, and production reliability. Hands-on experience implementing and operating Kubernetes platforms across AWS, GCP, Azure, VMware Tanzu, OpenShift, Talos Linux, and on-prem environments. Strong background in SLI/SLO practices, RCA, postmortems, MTTD/MTTR improvement, chaos engineering, Terraform/Packer/Ansible automation, GitHub Actions, Azure DevOps, CloudBees Jenkins, GitLab, ArgoCD, FluxCD, Prometheus, Grafana, Loki, Datadog, Sumo Logic, CloudWatch, JFrog migration, and CVE remediation workflows.

SCALE & SYSTEM DESIGN HIGHLIGHTS

- **Scale & Availability:** Operated and designed multi-region cloud infrastructure serving high-throughput SaaS workloads (>1,000 req/sec) with active-active high-availability (HA) topologies.
- **Cost Efficiency:** Audited and optimized Kubernetes resource allocation and cleaned up stale infrastructure, achieving a 20% reduction in Azure cloud spend.
- **Reliability Engineering:** Defined organization-wide SLI/SLO guardrails, championed blameless postmortem cultures, and built automated GitOps security remediation pipelines (reducing MTTR by ~30%).

CORE TECHNICAL SKILLS

- **SRE & Reliability:** Incident Leadership (Incident Commander), SLI/SLO implementation, error-budget policies, blameless postmortem culture, RCA, runbook automation, MTTD/MTTR reduction, production support, release governance.
- **Kubernetes & Platform:** Multi-tenant GitOps fleet architectures, AWS/EKS, GCP/GKE, Azure/AKS, VMware Tanzu, OpenShift, Talos Linux (immutable OS), Cilium eBPF network routing, MetalLB, Traefik, Helm, cluster capacity planning.
- **Infrastructure as Code:** Terraform, Packer, Terragrunt, reusable modules, cloud resource provisioning, remote state, Crossplane, Ansible, Bash, Python.
- **CI/CD & GitOps:** GitHub Actions, Azure DevOps, CloudBees Jenkins, GitLab CI, ArgoCD App-of-Apps, FluxCD fleet patterns, automated rollbacks, environment promotion gates.
- **Observability:** LGTM Stack (Loki, Grafana, Tempo, Mimir), VictoriaMetrics, Prometheus, Datadog APM, Sumo Logic, alert noise reduction, dashboard visualization, metric/log correlation.
- **Cloud & Platform:** AWS CloudWatch/EC2/RDS/Route53/VPC/EKS; GCP GKE/App Engine; Azure AKS/Key Vault/VMSS/CDN/DNS; VMware Tanzu and OpenShift platform operations.
- **Security & Compliance:** CVE triage, vulnerability remediation automation, SAST/SCA/DAST guardrails, image hardening, registry migration (DockerHub to JFrog), SOPS, secrets management, platform hardening.

PROFESSIONAL EXPERIENCE

Improving India, formerly InfraCloud Technologies - Senior Consultant 1 - Senior SRE

Sep 2024 - Present

Client engagements: Acquia, Beamer, Medtronic

Technology stack: AWS, GCP, VMware Tanzu, OpenShift, Kubernetes, Terraform, Packer, Ansible, Python, Dockerfiles, CloudBees Jenkins, GitHub Actions, GitLab, ArgoCD, Helm, Prometheus, Grafana, Loki, Sumo Logic, Velero, SOPS, JFrog, GitOps, ITSM.

PROJECT: ACQUIA - AWS CLOUD PLATFORM / DIGITAL EXPERIENCE RELIABILITY

Role: Senior Consultant 1 - Senior SRE

Scope: AWS infrastructure automation, Kubernetes/platform upgrades, CloudBees Jenkins automation, DockerHub to JFrog migration, QA automation acceleration, observability, and chaos engineering enhancement.

- Implemented and enhanced AWS infrastructure automation using Terraform, Packer, Ansible, Python, Dockerfiles, and CloudBees Jenkins to support repeatable infra/application updates across platform environments.
- Enhanced CloudBees Jenkins automation that updated both applications and infrastructure, including Kubernetes version upgrades, platform component upgrades, validation workflows, and controlled release execution.
- Supported Kubernetes platform operations on AWS-backed environments including version upgrade planning, namespace checks, RBAC-aware troubleshooting, workload rollouts, Helm validation, service/ingress checks, ConfigMap/Secret review, pod scheduling issues, and resource request/limit review.
- Worked on DockerHub to JFrog migration by updating Dockerfiles, image references, build/publish flows, registry authentication patterns, and validation steps so application images could move to the approved artifact registry.
- Built and improved automation scripts in Python/Ansible for infra update tasks, application deployment checks, environment validation, and operational repeatability.
- Enhanced QA test automation by integrating Selenium-based test execution into the delivery workflow, reducing test execution time from approximately 3 hours to 30-45 minutes.
- Implemented and supported SLI/SLO-style monitoring for AWS-hosted workloads by tracking deployment health, workload readiness, restart frequency, errors, latency symptoms, resource saturation, storage/I/O pressure, and platform health.
- Built and improved Grafana/CloudWatch-style operational dashboards for workload availability, deployment status, pod restarts, node utilization, storage symptoms, and service-level visibility.
- Participated in incident response, RCA, and post-incident improvement activities by reviewing logs, metrics, Kubernetes events, deployment history, Jenkins execution logs, and recent infrastructure changes.
- Enhanced existing chaos engineering practices by improving failure-scenario validation, recovery checks, and runbook coverage for platform/application resilience testing.
- Created and improved runbooks/MOPs for Kubernetes upgrades, infra updates, application releases, rollback validation, QA automation execution, incident triage, and cluster/application health verification.
- Supported CVE and image remediation workflows by reviewing impacted workloads, validating image/version upgrades, checking deployment manifests, and coordinating GitOps-compatible patching.
- **Outcome:** Reduced QA validation time from about 3 hours to 30-45 minutes, improved registry compliance through DockerHub to JFrog migration, strengthened AWS platform upgrade automation, and improved resilience validation through enhanced chaos engineering practices.

PROJECT: BEAMER - GCP APP ENGINE TO GKE MIGRATION AND SAAS RELIABILITY

Role: Senior Consultant 1 - Senior SRE

Scope: Migration from GCP App Engine to GKE, Helm packaging, GitHub Actions CI/CD, monitoring, alerting, and cloud budget visibility.

- Migrated 15 applications from GCP App Engine to GKE by containerizing workloads, creating Helm charts, defining Kubernetes Deployments/Services/Ingress configuration, and validating application behavior after migration.
- Implemented GitHub Actions automation for build, validation, image publishing, Helm-based deployment, environment promotion, and post-deployment checks across GCP environments.
- Converted App Engine deployment patterns into Kubernetes/GKE deployment models with environment-specific Helm values, resource requests/limits, readiness/liveness probes, rollout checks, and rollback-friendly release structure.

- Added monitoring and alerting for GKE workloads covering service health, pod restarts, deployment failures, resource saturation, error patterns, and application availability.
- Implemented budget and cost visibility practices for GCP infrastructure by reviewing workload sizing, cluster utilization, resource requests, and environment-level spend signals.
- Supported incident management by triaging production issues, reviewing GCP/Kubernetes logs and metrics, identifying probable root causes, coordinating escalation, and validating service recovery.
- Improved MTTD/MTTR by creating focused dashboards, actionable alerts, and repeatable troubleshooting paths for migrated GKE applications.
- Supported release management across lower and production environments by validating deployment status, checking error/log signals after release, and coordinating rollback or fix-forward decisions when required.
- **Outcome:** Migrated 15 applications from App Engine to GKE, improved deployment repeatability through Helm and GitHub Actions, added stronger monitoring/alerting, and improved GCP cost visibility for the SaaS platform.

PROJECT: MEDTRONIC - ON-PREM VMWARE TANZU AND OPENSIFT PLATFORM RELIABILITY

Role: Senior Consultant 1 - Senior SRE

Scope: Client infrastructure optimization, image hardening, application restructuring, Ansible/Terraform enhancements, provisioning on VMware Tanzu and OpenShift, GitLab-based delivery support, observability, and platform hardening.

- Optimized client on-prem platform infrastructure across VMware Tanzu and OpenShift by reviewing workload architecture, deployment structure, resource configuration, platform constraints, and operational bottlenecks.
- Worked on image hardening by reviewing base images, image versions, vulnerability exposure, Dockerfile structure, registry usage, and upgrade/remediation paths for application workloads.
- Supported application restructuring for Kubernetes readiness by reviewing deployment layout, configuration separation, resource requests/limits, probes, service exposure, and environment-specific configuration patterns.
- Enhanced Ansible and Terraform automation for repeatable infrastructure provisioning, configuration management, environment updates, and operational consistency across Tanzu/OpenShift-backed environments.
- Provisioned and validated infrastructure on VMware Tanzu clusters and OpenShift, including namespace/project setup, workload deployment, service routing, access/RBAC checks, and platform add-on verification.
- Supported GitLab-based workflows for code/configuration changes, deployment coordination, review flow, and release validation.
- Worked on workload resiliency by reviewing replica availability, readiness/liveness probes, HPA behavior, deployment strategy, failure symptoms, and rollback/fix-forward steps during releases and incidents.
- Implemented and improved observability views using Prometheus/Grafana/Loki/Sumo Logic-style logs and metrics for platform health, workload errors, pod restarts, node/resource saturation, and deployment status.
- Participated in incident management and RCA by correlating Kubernetes events, pod logs, metrics, release timelines, GitLab changes, and recent configuration updates to identify root cause and recovery actions.
- Created runbooks/MOPs for operational tasks, image hardening checks, infrastructure provisioning, release validation, incident triage, and platform recovery procedures.
- **Outcome:** Improved on-prem Tanzu/OpenShift platform reliability, strengthened image security posture, enhanced infrastructure automation, and improved release confidence through better provisioning, observability, and operational documentation.

Opcito Technologies - Senior DevOps Engineer

Oct 2022 - Sep 2024

Client engagement: EMTECH

Technology stack: Azure, AKS, Terraform, Azure DevOps, GitHub Actions, Datadog, Docker, Kubernetes, SAST/SCA/DAST, Key Vault, VMSS, CDN, DNS.

PROJECT: EMTECH - AZURE CENTRAL BANKING / CBDC INFRASTRUCTURE PLATFORM

Role: Senior DevOps Engineer

Scope: Azure infrastructure and application delivery, 14 AKS cluster deployments, Azure DevOps CI/CD, GitHub source integration, QA/dev automation, runbooks, HA design, cost optimization, and chaos engineering implementation.

- Deployed and maintained 14 AKS clusters using Azure DevOps-driven infrastructure workflows, supporting environment provisioning, application hosting, HA design, and production-style platform operations.
- Managed both infrastructure and application delivery responsibilities, connecting GitHub-hosted codebases with Azure DevOps pipelines for build, validation, deployment, database workflows, test execution, and release governance.
- Built extensive Azure DevOps CI/CD pipelines for application deployment, database data operations, environment validation, automated testing, release approvals, rollback/fix-forward workflows, and operational runbooks.
- Created Azure DevOps runbooks and automated workflows to reduce pressure on QA and development teams by standardizing repeated deployment, validation, testing, and data-handling activities.
- Used Terraform and Azure DevOps automation for Azure infrastructure provisioning and configuration management, including AKS components, networking dependencies, Key Vault integration, DNS/CDN-related configuration, and environment-specific infrastructure.
- Implemented Azure AKS platform operations including node pool sizing, workload rollouts, HPA/autoscaling review, ingress/service troubleshooting, namespace validation, ConfigMap/Secret review, deployment health checks, and cluster upgrade support.
- Personally implemented chaos engineering practices for EMTECH by defining failure scenarios, executing resilience tests, validating recovery behavior, and documenting gaps/runbooks for application and infrastructure recovery.
- Integrated Datadog APM, infrastructure metrics, logs, and monitors across application and platform layers to improve visibility into latency, errors, resource saturation, service health, and deployment impact.
- Performed Azure cost optimization by minimizing unnecessary node capacity, deleting stale infrastructure, reviewing environment usage, right-sizing workloads, and improving autoscaling/resource allocation.
- Implemented HA infrastructure patterns for critical environments, validating redundancy, recovery behavior, deployment safety, and operational readiness.
- Supported incident response and RCA by correlating Datadog traces/logs/metrics with Azure DevOps deployment history, infrastructure events, application changes, and test results.
- Implemented security and release guardrails including SAST, SCA, DAST, image scanning/version review, dependency checks, and controlled deployment gates.
- **Measurable outcomes:** Deployed 14 AKS clusters, reduced QA/dev operational load through Azure DevOps automation and runbooks, reduced security/performance incidents by approximately 40%, reduced Azure cloud spend by approximately 20% through right-sizing/stale-infra cleanup, and improved deployment consistency by approximately 30% through pipeline standardization.

Alepo Technologies - Senior Engineer R&D DevOps

Aug 2021 - Oct 2022

Technology stack: OpenShift, OpenStack, Kubernetes, Jenkins, Helm, Docker, jMeter, Linux, VNF/telecom workloads, log analysis.

PROJECT: SAMSUNG KOREA - OPENSIFT PLATFORM FOR TELECOM/5G VNF WORKLOADS

Role: Senior Engineer R&D DevOps

Scope: OpenShift deployment, Kubernetes platform delivery, CI/CD, Helm packaging, performance testing, log analysis, and RCA.

- Led OpenShift deployment on OpenStack for telecom workloads, including cluster preparation, platform validation, namespace/project setup, workload deployment checks, and operational readiness validation.
- Implemented Kubernetes/OpenShift workload deployment patterns using Helm charts, Kubernetes manifests, Services, Routes/Ingress equivalents, ConfigMaps, Secrets, probes, and resource configuration.
- Built Jenkins pipelines for VNF image build, package validation, Helm deployment, and environment-specific release workflows.
- Supported rolling deployment and release validation activities by checking rollout status, logs, pod health, service routing, and application readiness after deployment.
- Performed stress and reliability testing using jMeter to validate workload behavior under load, identify bottlenecks, and capture failure conditions before production release.
- Conducted log analysis and RCA for orchestration, scheduling, routing, tracing, and workload performance issues.
- Worked on high-throughput telecom workload considerations including infrastructure sizing, virtual function behavior, pod scheduling, and resource contention symptoms.
- Created operational documentation and troubleshooting notes for deployment failures, platform checks, and repeatable validation procedures.
- **Outcome:** Improved deployment agility and platform readiness for containerized telecom workloads while reducing release risk through Jenkins automation, Helm packaging, and load-test-driven validation.

Allot Communications - Professional Services DevOps Engineer

Nov 2019 - Jul 2021

Technology stack: Kubernetes HA, OpenStack, Ansible, Bash, Linux, Prometheus, Docker, telecom environments, on-prem infrastructure.

PROJECT: TELECOM KUBERNETES HA DEPLOYMENTS - PHILIPPINES, INDIA, UGANDA

Role: Professional Services DevOps Engineer

Scope: On-prem Kubernetes HA deployment, automation, customer-site implementation, incident/change support, monitoring, and production-adjacent operations.

- Deployed and supported on-prem Kubernetes HA clusters on OpenStack for telecom customer environments, including control plane/worker setup, networking validation, workload deployment, and operational handover.
- Automated infrastructure and application deployment activities using Ansible and Bash to reduce manual steps and improve repeatability across customer sites.
- Implemented Kubernetes operational tasks including namespace setup, workload deployment validation, service checks, ingress/routing troubleshooting, pod/node health checks, and resource review.
- Supported production-adjacent incidents by analyzing Linux system logs, Kubernetes pod logs, events, node status, service availability, and infrastructure symptoms.
- Created MOPs and runbooks for deployment, change execution, rollback steps, troubleshooting, and customer handover.
- Used Prometheus-based monitoring and Linux tooling to review infrastructure health, resource usage, and service symptoms during deployment and support windows.
- Coordinated change-management activities with customer teams, including pre-change validation, implementation, post-change testing, and issue escalation.
- **Outcome:** Improved repeatability of telecom platform deployments, reduced manual configuration risk, and supported stable operations for carrier-grade customer environments.

Earlier Experience - Linux Server Administrator

Jan 2015 - Apr 2019

Technology stack: Linux, Windows Server, AWS EC2/RDS/Route53/VPC, LAMP, pfSense, LDAP, VPN, DNS, backups.

PROJECT: SYSTEMS ADMINISTRATION, AWS MIGRATION, AND NETWORK OPERATIONS

Role: Linux / Systems Administrator

Scope: Server administration, cloud migration support, networking, backups, user access, and operational support.

- Managed Linux and Windows server environments including package management, service configuration, log review, backup checks, patching, and troubleshooting.
- Supported AWS migration and operations using EC2, RDS, Route53, VPC, security groups, and basic network/service configuration.
- Administered LAMP stack workloads, DNS, firewall/VPN access using pfSense, LDAP/user access, and operational support tickets.
- Automated recurring administrative tasks using shell scripting and documented repeatable operational procedures.
- **Outcome:** Built a strong foundation in Linux, networking, cloud operations, troubleshooting, and production support that later evolved into DevOps/SRE responsibilities.

SELECTED HANDS-ON PROJECTS & OPEN SOURCE

KubeWise - Kubernetes Cost & Performance Optimization

URL: <https://github.com/jamilshaikh07/kubewise>

Technology stack: Python, FastAPI, SQLAlchemy, SQLite, Kubernetes API, Next.js, Helm, Docker Compose, Prometheus-style metrics model.

- Implemented an advisory-only Kubernetes optimization platform that connects read-only to clusters, ingests workload/resource data, detects missing requests/limits and over-provisioned CPU/memory, estimates potential savings, and generates advisory YAML/kubectl patches.
- Built a FastAPI backend with REST APIs, SQLAlchemy models, demo data seeding, pricing configuration, recommendation logic, and API smoke/unit tests.
- Built a Next.js dashboard to show cluster inventory, recommendations, savings estimates, risk levels, filters, charts, and patch previews.
- Implemented a Python Kubernetes collector/agent with read-only RBAC to list nodes, namespaces, pods, deployments, resource specs, and cluster allocatable capacity.
- Designed safety controls so the platform never auto-applies changes; recommendations remain advisory and require human review.
- Packaged the system with Docker Compose and Helm charts for local demo and Kubernetes deployment.
- **Outcome:** Demonstrates hands-on Kubernetes resource optimization, recommendation-engine design, cost visibility, safe-change workflow, and platform product thinking.

Talos Proxmox GitOps - Production-Style Homelab Platform

URL: <https://github.com/jamilshaikh07/talos-proxmox-gitops>

Technology stack: Proxmox, Talos Linux, Kubernetes, Terraform, Ansible, ArgoCD, Helm, Cilium, MetalLB, Traefik, cert-manager, trust-manager, external-dns, Cloudflare Tunnel, VictoriaMetrics, Grafana, Uptime Kuma, Trivy Operator, GitHub Actions.

- Implemented a complete Infrastructure-as-Code and GitOps platform to deploy a Kubernetes homelab on Proxmox using a three-layer model: Terraform infrastructure provisioning, Ansible/Talos cluster bootstrap, and ArgoCD-managed platform applications.
- Built Terraform automation for Talos control-plane/worker VM provisioning, VM sizing, scale planning, deterministic MAC/IP planning, and infrastructure lifecycle operations.
- Implemented Ansible automation for Talos bootstrap, Cilium installation, kubeconfig/talosconfig handling, cluster health checks, and inventory regeneration.

- Implemented ArgoCD App-of-Apps architecture to manage platform add-ons including MetalLB, Traefik, cert-manager, trust-manager, Cilium day-2 config, CoreDNS k8s-gateway, external-dns, Cloudflared, VictoriaMetrics stack, Uptime Kuma, Trivy Operator, local-path-provisioner, and metrics-server.
- Implemented ingress and access patterns with Traefik, MetalLB VIPs, Cloudflare Tunnel, external-dns, internal DNS, and TLS automation.
- Implemented operational commands and runbooks via Make targets for full deployment, layer-by-layer deployment, health checks, scaling, status, cleanup, access setup, and troubleshooting.
- Implemented GitHub Actions workflow for deployment orchestration using a self-hosted runner and layer-level skip inputs.
- **Outcome:** Demonstrates full hands-on Kubernetes platform engineering, IaC, GitOps, ingress/DNS/TLS, observability, security scanning, and day-2 operations.

Self-Hosted PaaS / Vercel Clone

URL: <https://github.com/jamilshaikh07/vercel-clone>

Technology stack: Go, Kubernetes, Talos, GitHub App, HMAC webhooks, Kaniko, CloudNativePG/Postgres, Traefik IngressRoute, external-dns, Cloudflare Tunnel, ttl.sh registry.

- Built a self-hosted PaaS that turns GitHub pushes into live Kubernetes deployments on a Talos cluster.
- Implemented a Go control-plane service for webhook ingestion, HMAC validation, deployment dispatch, persistence, and operational endpoints.
- Implemented database-backed deployment tracking using CloudNativePG/Postgres with idempotent migrations and webhook audit logging.
- Implemented rootless in-cluster image builds using Kaniko Jobs and GitHub installation tokens for repository clone/build workflows.
- Implemented Kubernetes deployment automation that creates Deployment, Service, and Traefik IngressRoute resources for tenant applications.
- Integrated external-dns and Cloudflare Tunnel so generated application hostnames can become publicly reachable without direct port exposure.
- Implemented idempotency and stale-state reconciliation patterns so worker restarts can reattach to running build jobs and reduce stranded deployment records.
- **Outcome:** Demonstrates platform engineering, Kubernetes controllers/workflows, secure webhook handling, build automation, GitOps-style deployment concepts, and production-style service exposure.

autofix-dojo - DevSecOps Automation

URL: <https://github.com/jamilshaikh07/autofix-dojo>

Technology stack: Python, Typer, FastAPI, DefectDojo API, Kubernetes, Helm, ArgoCD manifests, GitHub/GitLab CLI, Prometheus metrics, Docker, CronJobs.

- Built a GitOps-native automation tool that connects to DefectDojo, identifies Critical/High vulnerabilities, suggests safe image version updates, modifies Kubernetes/Helm manifests, and creates pull requests for remediation.
- Implemented Helm chart scanning for ArgoCD Application manifests, chart version comparison, priority-based batching, and step-by-step upgrade PRs for major version jumps.
- Implemented CLI commands for scan-and-fix, dry-run mode, finding listing, SLO view, smoke tests, Helm scanning, Helm upgrade PR creation, and manual Kubernetes job triggering.
- Built a FastAPI web dashboard with quick actions, CronJob/job visibility, logs, health endpoint, and Prometheus metrics.
- Packaged deployment as a Helm chart with CronJobs, RBAC, ServiceMonitor, ingress, and environment-specific values.
- **Outcome:** Demonstrates hands-on DevSecOps, vulnerability remediation automation, GitOps PR workflows, Helm upgrade governance, observability, and measurable remediation tracking.

CERTIFICATIONS

- CKAD - Certified Kubernetes Application Developer
- AZ-400 - Microsoft Azure DevOps Engineer Expert
- AZ-104 - Microsoft Azure Administrator
- HashiCorp Terraform Associate

EDUCATION

- B.Sc. in Information Technology - Pillai's College, Mumbai (2014 - 2018)
- Diploma in Computer Engineering - DY Patil Polytechnic, Mumbai (2007 - 2013)